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Risky news and credit market sentiment ⋆

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ABSTRACT

The nonlinear nexus between financial conditions indicators and the conditional distribution of GDP growth has recently been challenged. We show how one can use textual economic news and word embeddings to construct an alternative indicator which by design associates growth-at-risk to news about financial conditions. We then document that the proposed indicator is particularly informative about the lower left tail of the GDP distribution and delivers significantly better out-of-sample density forecasts than commonly used alternatives. Speaking to theories on endogenous information choice and credit-market sentiment we further document that the news-based index likely carries information about beliefs rather than fundamentals.

1. Introduction

Macro-prudential policies, adopted by numerous institutions around the world, rely on economic models emphasizing the role of the financial sector in creating and propagating vulnerabilities in the economy (Minsky, 1977, 1986; Kindleberger, 1978; Gertler and Bernanke, 1989; Kiyotaki and Moore, 1997; Bernanke et al., 1999). From a time series perspective this theory and practice have spurred a large literature documenting how financial indicators of, e.g., credit, leverage, and funding, have a strong association with the downside risk, as opposed to the central tendency, of future output growth. The work by Giglio et al. (2016) and Adrian et al. (2019) have been particularly influential, showing how financial indicators combined with quantile regressions, or related non-linear methods, can be used to produce out-of-sample growth predictions with potentially fat tails capturing growth-at-risk (GaR).[1]

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[1] Following Adrian et al. (2022) GaR is typically defined as the value of GDP growth at the lower fifth percentile of the predicted growth distribution, conditional on an index of financial stress. Prasad et al. (2019) provides a description of the use of this framework at, e.g., the IMF, and Greenspan (2004) and Kilian and Manganelli (2008) show how forecasting can be looked upon as a risk-managing exercise. See, e.g., Loria et al. (2022), Brownlees and Souza (2021), Figueres and Jarocinski (2020), and Chernis et al. (2023) for recent contributions to the GaR literature.

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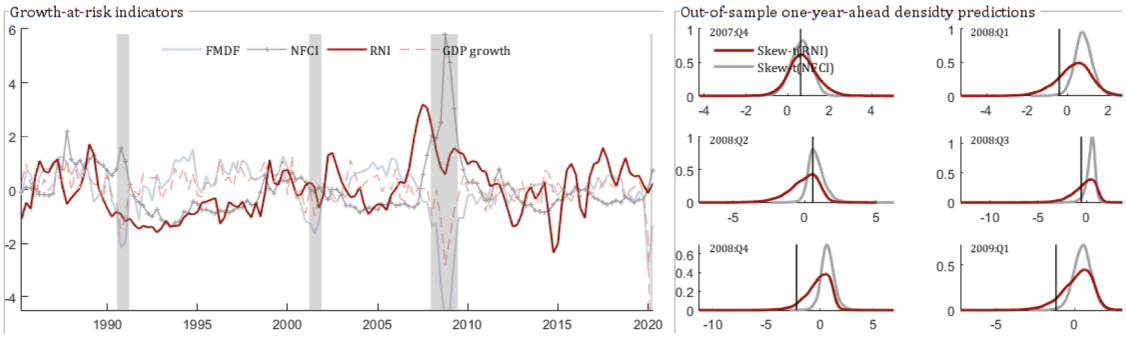


Fig. 1. The left graph reports the (standardized) estimates of quarterly U.S. GDP growth, and the monthly NFCI, RNI, and FMDF. The FMDF is the first common factor (FMDF) extracted from the FRED-MD dataset using a principal components decomposition. The observation dates are restricted by the data availability for estimating the RNI. Shaded areas represent NBER recession dates. The right graph reports the one-year-ahead out-of-sample density predictions obtained from fitting a Skew-t model to the data using either the NFCI or the RNI as explanatory variables for the time-varying moments. GDP growth outcomes are illustrated by the vertical black lines.

However, from a theoretical perspective it is still debated whether fluctuations in the "credit cycle" can best be understood as shocks to fundamentals coupled with, e.g., financial frictions, or with more behavioral mechanisms related to sentiment (Stein, 2021). Moreover, from an empirical perspective the nonlinear relationship between financial conditions indicators and the conditional distribution of GDP growth has not been uncontested either. For example, Hasenzagl et al. (2020) argue that the financial indicators usually considered, with the National Financial Condition Index (NFCI) in the U.S. being a leading example, do not provide robust and precise advance warnings about any features of the GDP growth distribution other than the mean. In particular, they argue, the NFCI contribute little to such distributional forecasts beyond the information contained in real indicators, such as the first common factor (FMDF) extracted from the FRED-MD dataset.[2]

Motivated by these debates and recent advancements in Natural Language Processing (NLP) we show how economic news combined with a shallow Neural Network can be used to construct an alternative financial indicator based on word embeddings. We then evaluate this index relative to the NFCI in terms of how well it characterizes and forecasts U.S. GDP growth using a parametric skewed Student-t (Skew-t) distribution which is commonly adopted in the recent GaR literature (Adrian et al., 2019; Hasenzagl et al., 2020; Delle Monache et al., 2023). Finally, we dissect the informational content of the derived index by linking it to shocks to either fundamentals or beliefs, where the latter speaks to popular sentiment-driven views on the credit cycle (Greenwood et al., 2016; Bordalo et al., 2018; López-Salido et al., 2017).

Our usage of word embeddings and economic news is motivated by the fact that news is timely, seldom subject to revisions, and might contain new information about economic fundamentals as well as fluctuations in sentiment (Larsen and Thorsrud, 2019). Similarly, word embeddings represent words in vector space and have been crucial for scientific advances and improvements on down stream tasks in the NLP literature over the last decade. They capture well shared context of words in the corpus, densely encode many linguistic regularities and patterns, and allow for arithmetic operations capturing associative meaning. Here we build on Mikolov et al. (2013), Caliskan et al. (2017), and Kozlowski et al. (2019), and take advantage of these properties by constructing monthly word embedding matrices based on a large corpus of business news provided by the *Dow Jones Newswires Archive* (DJ) covering the sample 1985M1 to 2020M4. At each point in time (month), we then project a word vector pointing in the growth-at-risk (recession risk) dimension in vector space on a word vector related to credit, leverage, and funding, and use the size of the association as a measure of financial conditions relevant for growth.

As seen in Fig. 1, the proposed indicator, which we label the Risky News Index (RNI), tends to increase at least one year prior to economic recessions. Compared to the NFCI and FMDF the RNI is more volatile but also seems to be leading. In line with this impression, we document using the Skew-t model that the RNI performs on-par with the NFCI for one-quarter-ahead predictions, but significantly improves upon it at the one-year-ahead horizon and during recessions periods. Fig. 1 provides an example, showing how the probability mass is shifted towards negative values to a much larger degree when using the RNI relative to the NFCI for GDP predictions during the Great Financial Crisis (GFC). Section 3 presents a battery of more formal tests of (density) forecast accuracy telling the same story: For the one-year-ahead horizon, and during recessions, the RNI provides significantly better predictions for the GDP growth distribution.

These results are important for at least three reasons. First, the recent debate about the nonlinear nexus between financial indicators and the distribution of future GDP growth questions the justification for macro-prudential policies based on GaR predictions (Hasenzagl et al., 2020). In contrast, our results are well in line with a large theoretical literature linking financial conditions to adverse growth outcomes and thereby also provide a more supportive view on such policies. Indeed, when performing an intrinsic evaluation of the estimated embeddings we find positive evidence regarding their financial interpretation and growth association.

[2] This dataset is compiled by McCracken and Ng (2016), contains over 100 monthly economic indicators, and builds upon the seminal contribution by Stock and Watson (1989), and the literature that followed, using large datasets for macroeconomic forecasting and monitoring.

Second, from a practical perspective informative one-year-ahead predictions are more valuable than shorter horizon predictions because it gives policymakers time to respond proactively on budding crises. For this reason, although we acknowledge that most studies in this literature, including ours, are riddled by relatively few observations and events, the suggested RNI is a potentially helpful alternative to commonly used indicators.[3]

Third, since it is implausible to assume that the RNI contains information about future shocks, its relatively strong predictive performance must be driven by properties of the underlying news data or by the technology used to construct it. In terms of the latter, we document that the RNI also outperforms simpler supervised NLP methods linking growth-at-risk to financial conditions, such as count- and Boolean-search-based methods commonly used in economics (Gentzkow et al., 2019; Ash and Hansen, 2023). As such, this study contributes to a growing field in economics documenting how news coupled with NLP can provide meaningful signals and predictions, see, e.g., Baker et al. (2016), Thorsrud (2020), Nyman et al. (2021), Angelico et al. (2022), and Adämmer et al. (2025) for applications measuring policy uncertainty, business cycle fluctuations, financial narratives and sentiment, inflation beliefs, and general macroeconomic risk, respectively. Thus far, however, word embeddings have not been widely applied although the methodology we propose easily can be adopted to other applications where capturing associations from written text is relevant. Still, we also find that the simpler NLP methods yield forecasting gains relative to the NFCI, suggesting an important role for the news itself.

To dissect the information content of the proposed index we proceed in three complementary directions. First, we use the Structural Vector Autoregression (SVAR) framework proposed by Enders et al. (2021) to estimate aggregate shocks to beliefs and fundamentals and analyze their correlation with the RNI. Second, we build on recent work by López-Salido et al. (2017) and investigate to what extent the RNI contains a predicable component related to lagged valuation indicators and expected returns on credit assets, which López-Salido et al. (2017) associate with credit market sentiment. Finally, we perform intrinsic evaluation exercises on the estimated word embeddings, asking whether the RNI aligns more closely with sentiment- or fundamental-laden language.

Our more structural analysis shows that the RNI is significantly correlated with belief shocks and not significantly correlated with shocks to fundamentals. Likewise, we find a significant correlation between lagged valuation indicators and the RNI. Both of these results support a view where the RNI captures shocks to expectations as in classic accounts of financial crises, emphasizing credit market sentiment (Minsky, 1977, 1986; Kindleberger, 1978). Consistent with this, using the underlying word embeddings we document that fluctuations in financial conditions and the RNI are more associated with sentiment-laden language than fundamentals.

The rest of this article is organized as follows: In Section 2 we describe the news data and the word embedding methodology. Section 3 introduces the Skew-t model and provides in- and out-of-sample evidence on the non-Gaussian news-GDP-growth relationship. Section 4 links the derived index to theories on endogenous information choice and credit-market sentiment, while Section 5 concludes.

2. News data and word embeddings

In the following we first present the DJ news corpus and the word embedding procedure. Then we describe how the news-based word embeddings are used to construct a measure of financial conditions relevant for growth. Finally, we provide more intuition for the word embedding methodology by conducting an intrinsic evaluation of the estimated embeddings.

2.1. Data and estimation

The DJ corpus consists of roughly 25 million news articles written in English, covering the period from 1985M1 to 2020M4. The database contains a large range of Dow Jones' news services, including content from *The Wall Street Journal*. This is *Dow Jones* company's flagship publication, and also one of the largest newspapers in the U.S. in terms of circulation. This means that it has a large footprint in both the U.S. and global media landscape and that important ongoing stories and discussions are well covered by this type of news outlet.

The news corpus is cleaned prior to estimation. We remove all email and web addresses, numbers, and special characters, erase punctuation, set all letters to lowercase, and remove words containing fewer than two or more than 15 letters. These feature selection steps reduce the vocabulary size to approximately 90,000 unique terms. The dimension reduction facilitates estimation and is common in the literature. Finally, for reasons made clear in the next section, the corpus is partitioned into monthly blocks of text.

The monthly data blocks are used to compute word embeddings. Word embeddings map a text corpus with p unique words to a collection of vectors $\boldsymbol{V} = (\boldsymbol{v}_1, \dots, \boldsymbol{v}_p)$ where each $\boldsymbol{v}_j \in \mathbb{R}^d$ for a prescribed embedding dimension d. While many different word embedding models exist, they can almost all be cast as the solution of an optimization

$$\arg\min_{\boldsymbol{U},\boldsymbol{V}} f(\boldsymbol{X}, \boldsymbol{U}\boldsymbol{V}) \tag{1}$$

for a particular corpus representation $\boldsymbol{X}$ and objective function f, where $\boldsymbol{U} = (\boldsymbol{u}_1, \dots, \boldsymbol{u}_n)'$ are vectors in $\mathbb{R}^n$ representing contexts, typically not of main interest. Here we apply the widely used word2vec algorithm (Mikolov et al., 2013), which is often implemented

[3] Related to this, while a large literature has highlighted the importance of "excessive" credit growth as an indicator of more long-run risk (Jordà et al., 2011, 2013; Reinhart and Rogoff, 2009; Schularick and Taylor, 2012), we also show that the RNI outperforms different credit cycle aggregates for one-year-ahead predictions.

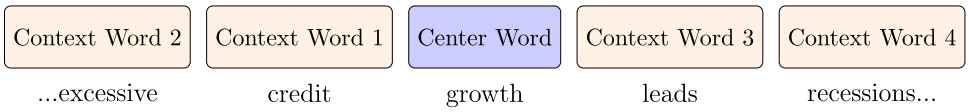


Fig. 2. Illustration of the word embedding methodology with context window $m = 5$: Predict context words given the center word.

as a skip-gram model with negative sampling.[4] For a given context window m, this method uses a binary classification problem, evaluating if the center word w_c is likely to show up in the context of the word w_j to compute the classifier weights that will be the actual word embeddings. For example, for a text sequence as illustrated in Fig. 2, the goal of the algorithm is to learn the word embedding representation for the word "growth" such that the probability of a context word such as "credit" occurring is high and a non-occurring word, such as "dog", is low.

In practice, the word2vec skip-gram model with negative sampling is solved using a two-layered neural network, where the appropriate vectors in $\boldsymbol{V}$ and $\boldsymbol{U}$ are selected in the first layer and their dot product computed in the second layer. Learning proceeds as follows: The corpus is read line-by-line in a sliding window of m words. At each iteration, a one hot input vector $\boldsymbol{x} \in \mathbb{R}^p$ of a center word selects $\boldsymbol{v}_c = \boldsymbol{V}\boldsymbol{x}$. Similarly, the jth context word embedding $\boldsymbol{u}_j$, for $j = 1, \dots, m$, for an actual context word is selected from $\boldsymbol{U}$ together with K noise terms, or embeddings $\boldsymbol{u}_{neg,k}$, representing words that are not occurring in the context window of the center word. Then, in the second layer word similarity is approximated by the dot product of the word and context embeddings and the sigmoid $\sigma(x)$ activation function is applied on the resulting scores. Finally, the model learns by maximizing the probability of true context words and minimizing the probability of negative samples using the loss function

$$L = -\log \sigma(\boldsymbol{u}_j' \boldsymbol{v}_c) - \sum_{k=1}^{K} \log \sigma(\boldsymbol{u}_{neg,k}' \boldsymbol{v}_c), \tag{2}$$

The word2vec algorithm with negative sampling is fast and available in many software packages. We set the context window $m = 5$, restrict the word embedding length to $d = 100$, and train the network using Stochastic Gradient Descent for five epochs on every monthly data partition.[5]

2.2. Constructing the RNI

In the NLP literature word embeddings are widely used because they encode linguistic regularities and patterns and allow for arithmetic operations. Here we take advantage of these properties and first compute two aggregate embeddings aimed at capturing "growth-at-risk" ($\boldsymbol{gr}_t$) and "financial conditions" ($\boldsymbol{fc}_t$). Next, we estimate how the similarity between these two embeddings changes across time. The leading hypothesis is that the size of the association is a useful measure of financial conditions relevant for future growth.

Eq. (3) defines the two aggregated vectors $\boldsymbol{gr}_t$ and $\boldsymbol{fc}_t$, where the individual terms on the right hand side of each equality sign are the word embeddings associated with that particular word in a given month t.

$$\begin{aligned} \boldsymbol{gr}_t &= \boldsymbol{recession}_t + \boldsymbol{risk}_t \\ \boldsymbol{fc}_t &= \boldsymbol{credit}_t + \boldsymbol{leverage}_t + \boldsymbol{funding}_t, \end{aligned} \tag{3}$$

The motivation for the $\boldsymbol{fc}_t$ aggregate is simply taken from the GaR literature, where financial conditions is linked to exactly fluctuations in credit, leverage, and funding. Indeed, the commonly used NFCI is a weighted average of financial indicators reflecting these categorizes.[6] Therefore, from (3) we obtain an aggregate vector aimed at pointing in the relevant financial dimension in the high-dimensional vector space. The motivation for the $\boldsymbol{gr}_t$ aggregate is somewhat more subtle. Since the term "growth" together with "risk" can be used in both a positive (upside risk) and negative (downside risk) context, but the GaR concept is explicitly defined to be associated with the left tail of the growth distribution (Adrian et al., 2022), it might be misleading to simply add $\boldsymbol{growth}_t$ to $\boldsymbol{risk}_t$ to capture growth-at-risk. Instead, we use the term "recession", which has a solely negative connotation, but test robustness to this choice in Section 3.3.

[4] The word2vec model can be trained using either the skip-gram or the continuous bag-of-words (CBOW) algorithm. Skip-gram works well with small datasets, and can better represent less frequent words. We therefore use this approach here. Other widely used word embedding models include GloVe (Pennington et al., 2014) and fastText (Joulin et al., 2016). Newer Transformer-based models, such as BERT (Devlin et al., 2018), explicitly learn context specific embeddings, and might potentially deliver higher quality embeddings. However, these models typically contain hundreds of millions parameters, making it unfeasible to estimate on a monthly basis. Transfer learning and fine-tuning approaches are possible, but would easily involve some type of forward-looking bias in our setting.

[5] In the original word2vec implementation $m = 8$ and $d = 300$. Because we estimate the model on a much smaller corpus, we also scale down the embedding and context window dimensions. We report robustness to these hyperparameter settings in Section 3.3. While word embeddings derived from neural networks are unknown to many economists, a potentially more familiar version would be to extract principal component estimates from the corpus' Pointwise Mutual Information (PMI) matrix, which contains the likelihood of words co-occurring in the corpus, and let the factor loadings represent the word embeddings. However, this simpler method enforces a more restrictive functional form.

[6] The NFCI, developed by Brave et al. (2012), is constructed as the common factor extracted from over 100 financial variables, covering four categories of data; credit quality, funding risk, nonfinancial and financial leverage.

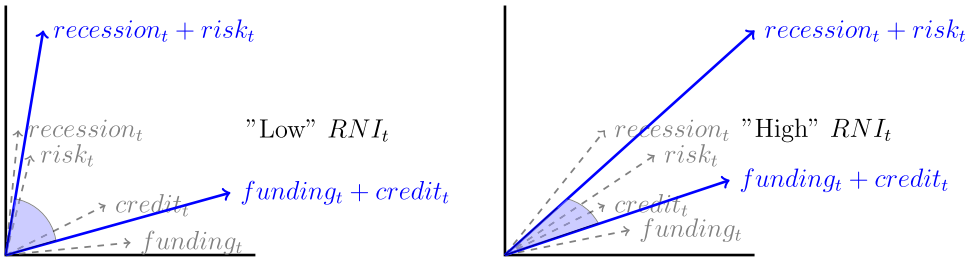


Fig. 3. RNI intuition and word embeddings illustrated in two dimensions. For visual clarity the vector for the term "leverage" is not included in the graphs.

To capture the monthly association between these two aggregates we follow the same intuition as in Kozlowski et al. (2019) and compute the cosine similarity between the two normalized $\boldsymbol{gr}_t$ and $\boldsymbol{fc}_t$ vectors such that

$$RNI_t = \frac{\boldsymbol{gr}_t \cdot \boldsymbol{fc}_t}{\|\boldsymbol{gr}_t\| \|\boldsymbol{fc}_t\|}. \tag{4}$$

This gives the same solution as linearly projecting $\boldsymbol{gr}_t$ onto $\boldsymbol{fc}_t$ and is also similar to the OLS solution. Thus, an increase in RNI_t implies a stronger association between how the news media writes about growth-at-risk and financial conditions.

Fig. 3 illustrates the RNI intuition when $d = 2$ and at two potentially different time periods. The RNI is low when the angle between the growth-at-risk and financial conditions vectors is large. In contrast, the RNI is high when these two vectors point in more or less the same direction. Naturally, words that are similar tend to cluster together in vector space. Importantly, however, because the optimal distance of two vectors is a function of shared context rather than co-occurrence, words need not co-occur in order for their vectors to be positioned close together. For example, if "recessions" and "credit" both appear near the word "risk" then the vectors for "recessions" and "credit" could be located near each other in the embedding space, even if they never appear together in the text. The RNI might, for example, increase because one set of articles writes more about recessions and risk than before while another set of articles writes more about financial conditions and risk. Although the two set of articles never make the connection between recession risk and financial conditions the RNI might because the embeddings capture the shared risk context. This makes the word embedding approach rather different than the traditional count- and Boolean-search-based methods often applied in economics. In Section 3.3 we document that the proposed approach also seems to perform better in the current context.

Earlier studies by, e.g., Caliskan et al. (2017), entertain the same type of geometrical-based intuition as illustrated in Fig. 3, but use word embeddings to show how machines can learn word associations from written texts and that these associations mirror those learned by humans. Similarly, Garg et al. (2018), Kozlowski et al. (2019), and Kapfhammer et al. (2020) use word embeddings to study changes in gender and ethnic stereotypes over time, changes in cultural associations, and climate change transition risk, respectively. Here we contribute by moving this type of reasoning into the realm of economics and growth-at-risk analysis.

In Section 2.3 we perform an intrinsic evaluation of the estimated embeddings and provide further intuition for the word embedding approach. Before that, we emphasize three additional aspects of the procedure. First, changes in the RNI_t across time can be due to either high-frequency changes in how the media relates growth-at-risk to financial conditions, more persistent changes in how this relationship is focused upon, or noise and breaks in the news coverage and style. To isolate the two former components we apply simple (backward-looking) moving average filters which first normalizes each observation with the mean and standard deviation of the last five years of raw data and then smooths the resulting series by the trailing six month average.[7]

Second, to take into account estimation uncertainty of the RNI we conduct subsampling. The approach is described in Appendix B.1. As seen in Fig. A.1 (Appendix A) the index is very precisely estimated. Going forward we only use the mean estimate. This treatment parallels common practice in the literature when employing other estimated indices such as the NFCI, which is likewise derived as a latent factor. While formal convergence results for factor-based estimators (e.g., Bai and Ng (2006)) support the use of such estimated indices in downstream analysis, analogous results for word-embedding-based measures are not yet available. Still, the empirical precision of the RNI suggests that any residual generated-regressor bias is unlikely to materially affect our findings.

Third, we are far from dogmatic about the chosen keywords in (3). If other researchers believe a different set of words would be more appropriate for linking growth risks to financial conditions there are no restrictions in the methodology preventing using such words. Related to this, from the GaR literature perspective, financial conditions in month t should be related to potentially adverse growth outcomes in the future, and not within the same month, as implied by (4). However, we do not interpret the growth-at-risk dimension literally as a GaR outcome, but a risk of this happening. I.e., the hypothesis we later test is whether a high correlation between news coverage of growth-at-risk and financial conditions *today* contains useful information for describing the conditional

[7] Similar types of normalization procedures are commonly applied in the news-based finance and economics literature (Tetlock et al., 2008; Baker et al., 2016; Caldara and Iacoviello, 2022). The length of the window used to remove the slow-moving trend component is motivated by the cycle lengths typically assumed in the literature studying debt and credit cycles, see, e.g., Hamilton and Leff (2020) and Drehmann and Yetman (2021) for recent applications and discussions regarding filtering methodology and Section 3.3 for robustness experiments regarding these choices.

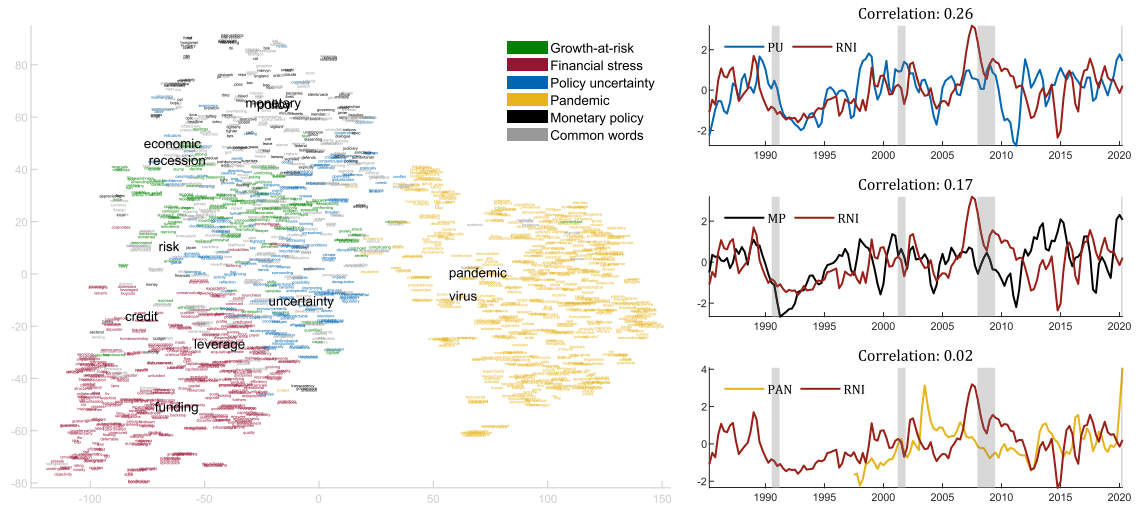


Fig. 4. The left graph reports a scatter plot of the words related to the five concepts listed in the legend. See the text and Appendix B.3 for a description of how the graph is constructed. The right graph reports the RNI together with the implied time-series for the three alternative concepts. The alternative time-series are computed using the same methodology as described in Section 2.2, but now replacing the $\boldsymbol{fc}_t$ vector by one of the alternatives defined in the text.

distribution of *future* GDP growth. Still, there is nothing in the methodology which prevents using a more dynamic specification in (4).[8]

2.3. The RNI in time and vector space

The mean estimate of the RNI is plotted together with the NFCI, FMDF, and quarterly GDP growth, in Fig. 1 in Section 1. The RNI tends to reach high levels at least one year prior to crisis episodes, but it is also somewhat volatile, at least compared to the NFCI. Moreover, while the FMDF and NFCI are highly correlated, particularly around crisis episodes, the RNI seems to be leading. In terms of explaining the conditional mean, this is more formally confirmed by the Granger-causality statistics presented in Fig. A.2 (Appendix A). The FMDF Granger-causes the NFCI but the NFCI does not Granger-cause the FMDF. The RNI, however, Granger-causes the other indicators whereas neither of the other series Granger-causes the RNI.[9]

Since word embeddings represent words as vectors, distance metrics, such as cosine similarity, are commonly used to perform intrinsic evaluation of the estimated embeddings. The word scatter plot in Fig. 4, produced using the t-SNE algorithm (Van der Maaten and Hinton, 2008), provides an example of this type of evaluation, where words colored red and green report the most similar words to the two aggregated vectors in (3). For ease of interpretability we focus on one particular year, 2007, and the average embeddings across months for this year. Figs. A.9 and A.10 (Appendix A), reports the 30 most related words in a more readable format and across six different years. For comparison and illustrative purposes, we also plot in Fig. 4 the most similar words to three alternative aggregate embeddings defined such that $\boldsymbol{mp}_t = \boldsymbol{monetary}_t + \boldsymbol{policy}_t$, $\boldsymbol{pan}_t = \boldsymbol{virus}_t + \boldsymbol{pandemic}_t$, and $\boldsymbol{pu}_t = \boldsymbol{economic}_t + \boldsymbol{policy}_t + \boldsymbol{uncertainty}_t$. Although we do not claim that these alternative constructions are the best possible approximations to the different concepts, they are distinct from the financial stress concept, and might all be important for understanding recession periods in our sample.

While one should be cautious about drawing strong conclusions regarding potential cluster sizes and distances in plots such as Fig. 4, the analysis delivers at least three relevant insights. First, words that are closely related to the words used to define the different concepts tend to cluster together. For example, both the "pandemic" and "financial conditions" terms form distinct clusters, signaling that these are very different concepts. The "monetary policy" and "policy uncertainty" terms, on the other hand, overlap to some extent because they are more related. Thus, although the concepts above share many words, they form distinct clusters because the aggregated embeddings capture associative meaning and shared context. Moreover, the "purity" of the clusters visualized in Fig. 4

[8] Since independent applications of the word2vec algorithm might result in arbitrary orthogonal transformations the estimated word embeddings must be aligned before they are compared and used across time. One commonly used method for this type of alignment is orthogonal Procrustes (Hamilton et al., 2016).

[9] The results presented in Fig. A.2 also speak to the critical review by Hasenzagl et al. (2020), who convincingly show that the information contained in the NFCI is encompassed by information already contained in real indicators, such as the FMDF. In particular, when calculating the two first principal components of the reduced form residuals from the VAR used to calculate the Granger causality statistics, we find that the FMDF, NFCI, and GDP load strongly on the first component whereas the RNI loads strongly on the second component. In other words, the NFCI and the FMDF seem to contain similar contemporaneous information, as argued in Hasenzagl et al. (2020), whereas the RNI is fairly exogenous to contemporaneous innovations in both the FMDF and NFCI.

are not a result of the t-SNE algorithm. Using K-means clustering to estimate clusters from the combined embedding matrix shows that the unique concept terms in Fig. 4 typically are allocated to distinct clusters (Table A.8 in Appendix A).

Second, as seen from Figs. A.9 and A.10, the most similar words to the aggregate embeddings in (3) are fairly stable across time. In addition, these wordclouds suggest that the aggregated vectors indeed point in the intended directions in vector space. E.g., among the most similar embeddings to the $\boldsymbol{gr}_t$ vector in 2006 are words such as "vulnerable", "slowdown", and "downturn".

Third, the results reported in the Fig. 4 might suggest that "growth-at-risk" is more related to "monetary policy" and "policy uncertainty" than "financial conditions". This can be true. A number of studies have highlighted the role of (systematic) monetary policy and (general) uncertainty for growth (see, e.g., Berger et al., 2020; Caldara et al., 2016; Jurado et al., 2015; Brunnermeier et al., 2021). However, the question we address is not how to best predict the conditional distribution of GDP growth, but, in light of theories linking financial conditions to growth vulnerabilities, whether an alternative measure of financial conditions adds value relative to commonly used indicators, such as the NFCI.

That said, one cannot rule out that news coverage of, e.g., monetary policy or (general) uncertainty, feeds into the RNI which in turn predicts GaR. If this was the case, however, we would expect to see a strong correlation between movements in the RNI and time series analogs of the alternative concepts. The right graph in Fig. 4 suggests that this is not the case. Although we observe co-movement in certain time periods, the overall correlation is relatively low. For the "pandemic" time series this finding is particularly clear. It starts late because of missing data, spikes around the SARS and COVID-19 pandemics in 2003 and 2020, but has very little overall correlation with the RNI. Moreover, when doing the out-of-sample forecasting experiment, described in Section 3.2, using the alternative "monetary policy" and "policy uncertainty" indicators, we find that they are outperformed by the Skew-t(RNI) model (Tables A.3 and (A.4 in Appendix A).

3. GDP and risky news

For characterizing and predicting the conditional GDP distribution we use a parametric Skew-t distribution. This framework is motivated by the influential work of Adrian et al. (2019), and has been widely adopted in the GaR literature, including in the more critical analysis of Hasenzagl et al. (2020). In particular, we follow Delle Monache et al. (2023), who extend the Skew-t framework by allowing for permanent and transitory changes of the distribution of GDP growth, and document that their specification produces competitive out-of-sample (point, density and tail) forecasts that improve upon standard benchmarks, and is essential to recover well-known facts, such as the Great Moderation (McConnell and Perez-Quiros, 2000), the fall in long-run growth (Antolin-Diaz et al., 2017), and negative and decreasing business cycle skewness over the last 30 years (Jensen et al., 2020).

More formally, we assume that quarterly (log) real GDP growth (Δy_t) can be written as

$$\Delta y_t = \mu_t + \upsilon_t, \quad \upsilon_t \sim Skt(0, \sigma_t, \alpha_t, \nu), \tag{5}$$

and the Skew-t specification follows Arellano-Valle et al. (2005). The location (μ_t), scale (σ_t), shape (α_t), and degrees-of-freedom (ν) parameters define the distribution's conditional mean, variance, asymmetry (skewness), and the thickness of the tails. Here, these parameters are allowed to be time varying, and, following Delle Monache et al. (2023) and Labonne (2025), evolve according to score-driven processes containing a persistent trend component and a transitory component.[10] Using link functions, such that $\gamma = log(\sigma)$ and $\varrho = arctanh(\alpha)$, and letting $u_t \in \{\mu_t, \gamma_t, \varrho_t\}$, the sum of the two components is $u_t = \tau_{u,t} + \upsilon_{u,t}$ with

$$\tau_{u,t} = \tau_{u,t-1} + \varsigma_u s_{u,t} \quad \upsilon_{u,t} = \phi_u \upsilon_{u,t-1} + \beta_u x_{t-1} + \kappa_u s_{u,t}, \tag{6}$$

where x is an exogenous predictor, e.g., the quarterly RNI or NFCI, and ς_u and κ_u denote the gain parameters giving weight to the time-varying scores $s_{u,t}$ of the predictive likelihood. Thus, parameter updates are defined by the steepest ascent direction for improving the model's local fit, and the trajectories of the time-varying parameters are perfectly predictable given past information.

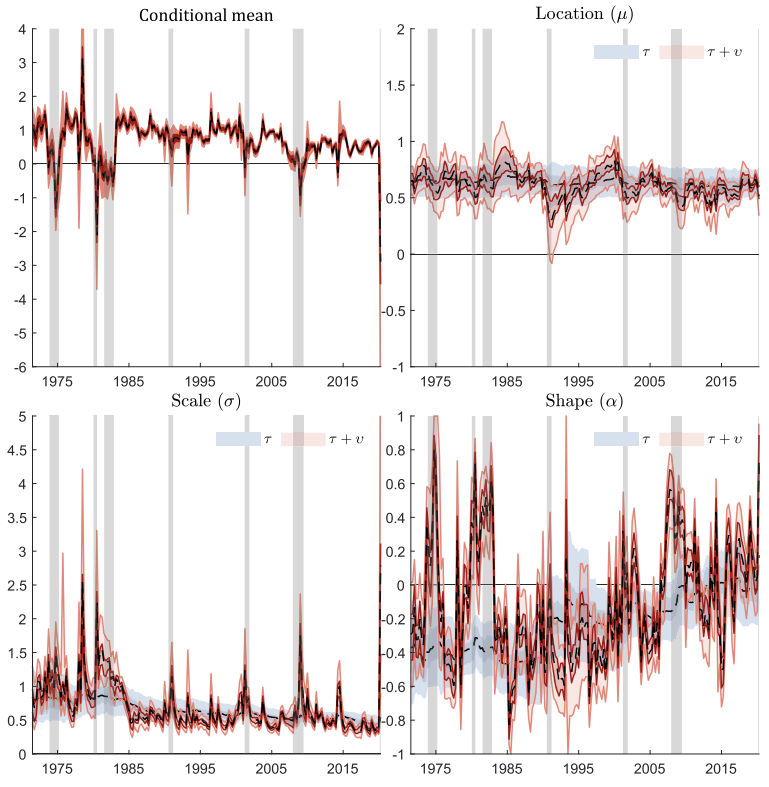
The model's unknown parameters are estimated using maximum likelihood (ML) where the log density at time t takes the form

$$\log p(\Delta y_t | Y_{t-1}) = \log C(\eta) - \log \sigma_t^2 - \frac{\eta+1}{2\eta} \log\left[1 + \frac{\eta(\Delta y_t - \mu_t)^2}{(1 - sgn(\Delta y_t - \mu_t)\alpha_t)^2 \sigma_t^2}\right], \tag{7}$$

with $\sigma_t > 0$, $\alpha_t = [-1, 1]$, $\eta = 1/\nu$, with $\nu > 1$, and $C(\eta) = \sqrt{\eta}\Gamma((\eta+1)/2\eta)/(\sqrt{\pi}\Gamma(1/2\eta))$ where $\Gamma(.)$ is the Gamma function. The distribution is skewed towards positive values if $\alpha_t < 0$ and towards negative values if $\alpha_t > 0$. When the tail parameters are very large and the shape parameter set to zero the distribution is equivalent to a normal distribution. To ensure that the transitory component in (6) does not contain a unit root, we restrict $|\phi_u| < 1$ during estimation.

As shown in Delle Monache et al. (2023), using the NFCI and the model specification in (5) and (6) produce significantly better out-of-sample predictions for U.S. GDP growth than those obtained in the original work by Adrian et al. (2019) and even the more

[10] In the interest of preserving space we refer to Delle Monache et al. (2023) and Labonne (2025) for detailed expositions of this model class and specification. We note here that we depart (slightly) from the modeling choices used in Delle Monache et al. (2023) along two dimensions. These are described in Appendix B.2. In general, score driven methods, introduced by Creal et al. (2013) and Harvey (2013), provide an efficient way of modeling dynamic distributions by using the conditional score of the likelihood as the driving force for the model's time-varying parameters. The likelihood of score driven models are generally available in closed form because they are observation-driven, making maximum likelihood an efficient estimation approach (Blasques et al., 2022). Koopman et al. (2016) show that they provide comparable forecasting performances to nonlinear non-Gaussian state space models. See Delle Monache and Petrella (2017), Gorgi et al. (2019), Delle Monache et al. (2023), and Labonne (2025) for recent forecasting applications.



Parameter estimates

	Skew-t(RNI)	Skew-t(NFCI)
ϕ_μ	0.827 (0.096)	-0.373 (0.486)
ϕ_γ	0.546 (0.129)	0.549 (0.237)
ϕ_ϱ	0.161 (0.139)	0.084 (0.199)
ς_μ	0.000 (0.000)	0.000 (0.000)
κ_μ	0.053 (0.036)	0.000 (0.000)
ς_γ	0.028 (0.018)	0.000 (0.000)
κ_γ	0.251 (0.058)	0.124 (0.081)
ς_ϱ	0.034 (0.015)	0.046 (0.018)
κ_ϱ	0.135 (0.050)	0.096 (0.056)
β_μ	0.056 (0.031)	-0.232 (0.123)
β_γ	0.089 (0.049)	0.128 (0.078)
β_ϱ	0.502 (0.127)	0.242 (0.106)
T	196	196
LL	-187.532	-180.138

Model specification tests

H0:	Skew-t(RNI)	Skew-t(NFCI)
α	-0.161 (0.170)	1.161 (0.170)

Fig. 5. The graphs report the estimated time-varying moments, together with 50% and 95% confidence bands, obtained from the Skew-t(RNI) model. The table reports the parameter estimates from both the Skew-t(RNI) and Skew-t(NFCI) models. Values in parenthesis are standard errors. Model specification tests for non-nested models with *H0: Skew-t(RNI)* versus *H1: Skew-t(NFCI)* are estimated following Davidson and MacKinnon (1981) from the regression $\Delta y_t - \hat{f}_t = \alpha(\hat{q}_t - \hat{f}_t) + \epsilon_t$, where $\hat{f}_t$ and $\hat{q}_t$ are the fitted values from the Skew-t(RNI) and Skew-t(NFCI) models, respectively. If *H0* is true, then the true value of α is zero. Alternatively, testing *H1* implies reversing the roles of *H0* and *H1*.

critical review by Hasenzagl et al. (2020). Accordingly, using (5) and (6) together with the NFCI is a natural and strong benchmark for assessing the potential value added of the RNI.

In the following we present in- and out-of-sample results using the Skew-t model in conjuncture with either the RNI or the NFCI. We then present robustness results and discuss how the word embedding approach adds value relative to related NLP approaches in this context.

3.1. In-sample evidence

Fig. 5 summarizes how the Skew-t(RNI) model characterizes GDP growth in-sample. To be able to compare our estimates with findings in the existing literature, such as Adrian et al. (2019) and Delle Monache et al. (2023), the model is estimated using data from 1973:Q3 to 2020:Q2. Since the RNI is only available from 1985:Q1, we construct synthetic RNI observations using its correlation with the NFCI, which is available for the whole sample. We do so by first regressing the RNI on eight leads and lags of the NFCI, and then use the resulting parameter estimates together with the NFCI to construct the values for the missing RNI history.

As seen in the figure, the time-varying moments are rather precisely estimated and vary significantly over time. Economic expansions tend to be characterized by positive skewness, and there is a positive correlation between the mean and variance of the conditional GDP distribution during such expansions. In contrast, during recessions, the correlation between the mean and variance turns negative. While these results are not new, and very similar to those from the Skew-t(NFCI) model in Delle Monache et al. (2023), they continue to hold when using the RNI as the exogenous variable in the model.[11]

The contribution of the RNI index to these time-varying patterns is illustrated in the table to the right in Fig. 5. An increase in the RNI does not significantly affect the location or scale of the GDP distribution, but significantly increases the asymmetry of the distribution. Moreover, the persistence of the transitory component is large and significant for the location, but insignificant for the shape, while the gain estimates are not significant for the location processes, but generally significant for both the scale and

[11] See, e.g., Smith and Vahey (2016), Salgado et al. (2019), Carriero et al. (2024), and Fernández-Villaverde and Guerrón-Quintana (2020) for related findings about how the correlations between higher order moments vary with the business cycle.

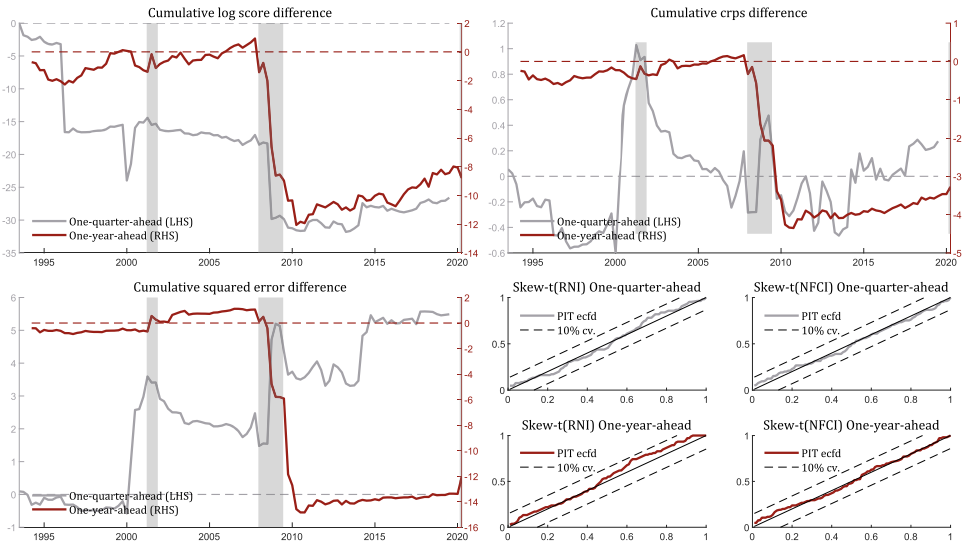


Fig. 6. The figure reports cumulative differences in LS, CRPS, and squared forecast errors between the Skew-t(RNI) and Skew-t(NFCI) models for one-quarter- and one-year-ahead predictions. All relative scores are normalized such that an increase implies a relative improvement of the Skew-t(NFCI) model. The lower right quadrant reports the empirical cumulative distribution function of the PITs together with the Kolmogorov-Smirnov class of test statistic proposed by Rossi and Sekhposyan (2019).

shape processes. One implication of this is that the trend component of the location process is close to constant, whereas the trend components of the shape and scale processes gradually declines and increases, respectively, over the sample.

Parameter estimates from the Skew-t(NFCI) model are reported in the second column in Fig. 5. Compared to using the RNI the NFCI contributes more towards describing the location of the GDP distribution and the persistence of the transitory component for this moment is small and insignificant. Model specification tests in the tradition of Davidson and MacKinnon (1981) are reported in the lower part of the table. Focusing on the models' conditional mean estimates, these results suggest that a null hypothesis where the Skew-t(NFCI) model is the truth versus the Skew-t(RNI) alternative can be rejected. However, when reversing the roles of the null and alternative hypothesis we find that we cannot reject that the Skew-t(RNI) model is true.[12]

3.2. Out-of-sample evidence

To evaluate the out-of-sample performance of the Skew-t(RNI) model we perform a quasi-real-time forecasting experiment, and compare its performance with the Skew-t(NFCI) model. The models are first estimated on the sample 1973:Q2 to 1993:Q1, and predictions are constructed for horizons one-quarter- to one-year-ahead. This process is then repeated for the remaining sample using an expanding estimation window.

For each quarterly forecasting vintage we assume that both the RNI and NFCI are available (at the last day of the quarter). Thus, one-quarter-ahead predictions are obtained directly from the model and the implied parameter estimates. For longer horizons forecasts we make the same assumption as in Delle Monache et al. (2023) and condition the predictions on the last available predictor values, i.e., assume their future values are approximated well by a random walk prediction. For multi-step ahead forecasting horizons the time-varying parameters are subject to additional uncertainty coming from the unobserved scores. To integrate this uncertainty into our forecasts we use the "bootcasting" algorithm of Koopman et al. (2017) which consists of sampling unobserved (scaled) scores from their in-sample realizations (see also Delle Monache et al., 2023).

To evaluate the predictions we use three different scoring metrics; the log-score (LS), the Continuously Ranked Probability Score (CRPS), and Root Mean Forecasting Errors (RMSE). To test for significant differences in predictive performance we use the conditional Giacomini and White (2006) test. We also evaluate the calibration of the predictive densities using the Probability Integral Transforms (PITs) and the test suggested by Rossi and Sekhposyan (2019).

Fig. 6 and Table A.1 (Appendix A), summarize our main results. The figure reports the PITs and the cumulative differences in LS, CRPS, and squared forecast errors between the Skew-t(RNI) and Skew-t(NFCI) models for one-quarter- and one-year-ahead predictions. All relative scores are normalized such that an increase implies a relative improvement of the Skew-t(NFCI) model. The Skew-t(NFCI) model is better on the one-quarter-ahead horizon for CRPS and RMSE loss, while the Skew-t(RNI) model is better according to LS. However, none of these differences are significant when considering the whole out-of-sample period, or when focusing on recession periods or the GFC in particular (Table A.1). In contrast, at the one-year-ahead horizon the Skew-t(RNI) model clearly outperforms

[12] Although the variance of the $\hat{\alpha}$ is biased in this simple setup, Davidson and MacKinnon (1981) argue that a crude test based on these types of regression will still be valid in the sense that the true (asymptotic) probability of Type I error will be no greater than the size of the test.

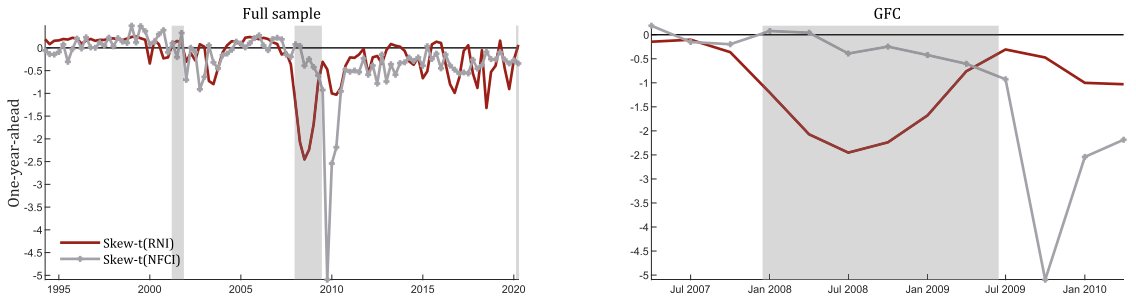


Fig. 7. The figure reports the one-year-ahead GaR predictions from the Skew-t(RNI) and Skew-t(NFCI) models. The figure to the left shows the predicted GaR for the whole out-of-sample evaluation periods while the figure to the right focuses on the GFC period.

the Skew-t(NFCI) model. For the RMSE loss the performance gain is roughly 10% when considering the full sample. Most of this gain is driven by recession periods and the GFC, and when focusing only on such episodes the differences in predictive performance are also generally significant at the 5% level. For completeness, Table A.2 (Appendix A), shows that these results are qualitatively similar if we instead compare the Skew-t(RNI) model to a Skew-t(FMDF) model.

The lower right quadrant of Fig. 6 reports whether the conditional predictive densities are correctly specified using the PITs and the Kolmogorov-Smirnov class of test statistic proposed by Rossi and Sekhposyan (2019). Under the null hypothesis the predictive density is correctly specified. In this case the empirical cumulative distribution function of the PITs should follow closely the 45 ° line. As seen in the figure, there are signs that the Skew-t(RNI) model has too many realizations falling in the center of the distribution at the one-year-ahead horizon (the slope in the center is steep relative to the 45 ° line). However, for both models and forecasting horizons, we can not reject the null hypothesis of correctly specified predictive densities.

The strong performance of the Skew-t(RNI) specification, particularly during recessions and the GFC episode, is not only statistically significant but also economically relevant. As seen from the density plots in Fig. 1 in Section 1, in every quarter of 2008 the Skew-t(RNI) model predicted considerably larger down-side risks than the Skew-t(NFCI) model. This stands in sharp contrast to similar plots in Hasenzagl et al. (2020), where there are essentially no differences between predictions based on either real or financial variables. An alternative illustration of this is given in Fig. 7, which reports the predicted GaR across the whole out-of-sample evaluation period and the GFC in particular. For example, one year prior to 2008:Q3, the Skew-t(RNI) model predicted that growth at the lower fifth percentile of the growth distribution would be -2.5 whereas the Skew-t(NFCI) prediction hardly had turned negative.

3.3. Robustness and alternative methodologies

The out-of-sample forecasting results reported in the previous section are robust along a number of different dimensions. First, in the experiment we use the final vintages of GDP and the NFCI. However, Amburgey and McCracken (2023) show that using the real-time estimates of the NFCI rather than the final vintage actually performs better in terms of predicting GaR. While the RNI is not subject to historical revisions, Table A.7 (Appendix A), shows that re-doing the experiment using real-time vintages of both GDP and NFCI does not alter our main conclusions.

Second, as described in Section 3.1 we extrapolate the RNI backward in time to estimate the Skew-t(RNI) on the same sample as the Skew-t(NFCI) model. Fig. A.3 (Appendix A) shows that our main conclusions regarding the forecasting performance of the Skew-t(RNI) relative to the Skew-t(NFCI) model, reported in Section 3.2, remain robust to starting estimation in 1985. We do, however, find that the in-sample characteristics of GDP growth using the Skew-t models deteriorate when a shorter sample is used. Thus, the high variance and crisis episodes in the 1970s and 1980s seem important when fitting the Skew-t model to the data.

Third, the smoothing operations used to construct the RNI are admittedly somewhat arbitrary. One could treat these choices as tuning parameters for minimizing loss on the downstream out-of-sample predictive task. We have not explored this alternative and leave that for future research. However, Fig. A.4 (Appendix A) shows that the out-of-sample forecasting results obtained in Section 3 are robust to using either the raw version of the RNI or a version without any smoothing.

Finally, we assess robustness by using the RNI for forecasting when it is derived from word embeddings estimated under alternative hyperparameter choices - either an embedding dimension of $d = 50$ or a context window of $m = 10$ - rather than the benchmark configuration $d = 100$ and $m = 5$. In addition, as a third alternative, we retain the benchmark embeddings but recompute the RNI using a $\boldsymbol{gr}_t$ vector defined as $\boldsymbol{growth}_t + \boldsymbol{risk}_t$, which is perhaps more literary related to growth-at-risk than our benchmark from (3). As shown in Fig. A.5 (Appendix A), the main results are largely robust to both hyperparameter settings and the precise definition of growth-at-risk: the RNI-based predictors tend to outperform the NFCI at the one-year horizon, but not at shorter horizons. At the same time, we observe a general deterioration of log-score performance relative to the benchmark predictions, and the results suggest that replacing “recession” with “growth” in the construction of $\boldsymbol{gr}_t$, or allowing for a larger context window, weakens the informational content of the resulting index.

In terms of applying alternative text-based methods we note that the word embedding methodology is supervised along two dimensions. First, the RNI is constructed by minimizing projection errors between a financial-conditions vector and a growth-at-risk vector. Second, the keyword set is deliberately selected, guided by economic theory linking financial conditions to growth-at-risk.

This latter feature aligns our approach with common count- and Boolean-search-based methods in economics (Gentzkow et al., 2019; Ash and Hansen, 2023). Unlike these methods, however, word embeddings incorporate supervised information at the estimation stage, which ensures that associations between words reflect their underlying economic relationships rather than artifacts of the unit of observation. For example, even if two relevant terms never appear in the same article, embeddings can still place them close in vector space if they share contextual usage.[13] Finally, representing words as vectors enables standard regression tools to quantify relationships between concepts - precisely the strategy employed here.

It is of course an empirical question whether the word embedding methodology adds value. For this reason we re-do the out-of-sample experiment for both a Skew-t(Count) and Skew-t(Bol) model. The alternative "Count" and "Bol" indicators are constructed using standard procedures and the same keywords defined in Section 2.2. In the interest of preserving space, a detailed description of how this is done is relegated to Appendix B.4. Fig. A.6 (Appendix A) reports the alternative indexes together with the RNI, and Tables A.5 and A.6 report the out-of-sample results. In short, both the "Count" and "Bol" indicators tend to spike around recession periods and is at times highly correlated with the RNI. In terms of out-of-sample performance we find no significant differences at the one-quarter ahead horizon. However, as before, the Skew-t(RNI) model tend to provide significantly better predictions than both the Skew-t(Count) and Skew-t(Bol) models at the one-year-ahead horizon, especially when considering recession periods and the GFC.

Still, although the Skew-t(RNI) model outperforms the two news-based alternatives, Figs. A.7 and A.8 (Appendix A) illustrate that using the alternative news-based indicators deliver predictions that for some horizons and score metrics are competitive, or even better, than those produced using the Skew-t(NFCI) model. This suggests that the informational content of the news plays an important role. In the next section we put further light on the mechanisms that might give rise to this finding.

4. Why is the RNI informative?

From a theoretical perspective there are at least two views linking financial conditions to economic downturns. In the classical theories motivating much of the growth vulnerability literature, borrowers and lenders are typically seen as fully rational but subject to various forms of credit limits or collateral constraints (Gertler and Bernanke, 1989; Kiyotaki and Moore, 1997; Bernanke et al., 1999). In turn, these financial market frictions play a central role in propagating and amplifying shocks to the economy, making economic downturns preceded by excessive credit growth and leverage especially severe. In essence, credit booms go bust when unexpected (exogenous) shocks happen (Jordà et al., 2011, 2013; Reinhart and Rogoff, 2009; Schularick and Taylor, 2012).[14]

An alternative literature emphasizes the role of sentiment for the credit cycle. This view draws on recent work in behavioral finance and on classic accounts of financial crises by Minsky (1977, 1986) and Kindleberger (1978), and asserts that predictable time variation in the sentiment of credit-market investors, due to, e.g., "diagnostic expectations" and extrapolative beliefs of investors, is an important determinant of the credit cycle (Greenwood et al., 2016; Bordalo et al., 2018). The empirical work by López-Salido et al. (2017) connects this to the business cycle and associates credit booms not with balance-sheet measures of credit growth but with proxies for the expected returns on credit assets. They then show that the economy performs poorly following periods when these proxies are unusually low by historical standards, i.e., when buoyant sentiment unwinds. In contrast to the balance-sheet-based view, no exogenous shock is needed. Once asset prices are significantly elevated an economic correction is closer at hand because overly optimistic investors will be disappointed going forward.

In principle, the RNI can inform both perspectives: it may reflect shifts in fundamental information as well as movements in sentiment and beliefs. Therefore, to better understand the informational content of the RNI and the mechanisms underlying its strong predictive power, we pursue three complementary approaches. First, building on Enders et al. (2021), henceforth EKM, we estimate time series capturing unexpected fundamental information and belief revisions, and examine their correlation with the RNI. Second, we construct credit-market sentiment measures using valuation indicators and the methodology of López-Salido et al. (2017), henceforth LSZ, tailored to the quarterly data and sample used here, and relate these measures to both the business cycle and the RNI. Finally, we return to intrinsic evaluation exercises, now asking whether the RNI aligns more closely with sentiment- or fundamental-laden language.

4.1. SVAR evidence

EKM propose a simple, but powerful, SVAR methodology for identifying aggregate shocks to expectations (belief shocks) from other fundamental shocks. Their basic premise is that any fluctuations in the (ex-post available) nowcast error that cannot be explained by new fundamental information has to be attributed to changes in beliefs. And, in turn, such belief shocks move output and the nowcast error in opposite directions. I.e., waves of optimism (or pessimism) have causal effects on economic outcomes. Below we replicate their analysis for the data and sample considered here.

More formally, let $E(\Delta y_t|I_t)$ denote the (median) nowcast of real GDP growth obtained from the Survey of Professional Forecasters (SPF) and define $ne_t = \Delta y_t - E(\Delta y_t|I_t)$ as the nowcast error. Then, following EKM, the VAR we consider can be written as:

$$\boldsymbol{y}_t = \boldsymbol{\Phi}\boldsymbol{x}_t + \boldsymbol{\varepsilon}_t \quad \boldsymbol{\varepsilon}_t \sim i.i.d.N(0, \boldsymbol{\Sigma}_\varepsilon), \tag{8}$$

[13] This arises because embeddings are estimated in a semi-supervised manner: running text provides an implicit supervision signal, eliminating the need for hand-labeled data.

[14] Although this literature mainly has focused on more long-run growth developments, we show in Appendix C that using the RNI also outperforms the credit-to-GDP gap for the out-of-sample growth predictions considered in Section 3.2.

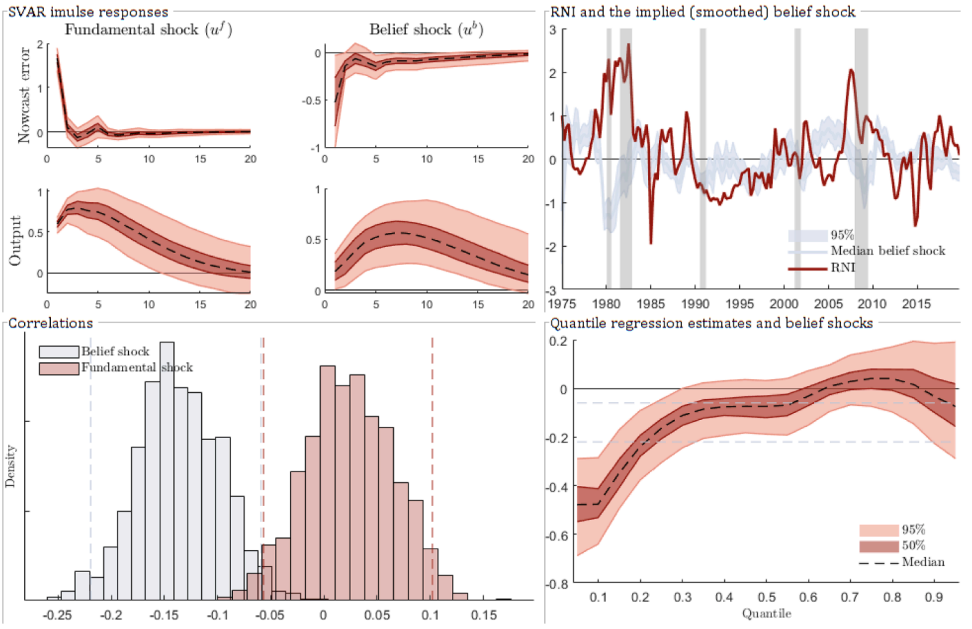


Fig. 8. The upper left graphs report the impulse responses, medians with posterior uncertainty, to a one standard deviation fundamental or belief shock. The y-axis denotes the response (in percent) and the x-axis the response horizon. The upper right graph reports the RNI together with the posterior distribution of u_t^b. For visual clarity, the belief shocks are smoothed using a three quarter moving average filter. The lower left graph reports the correlation coefficient between the RNI and (every draw) of either the (not smoothed) u_t^b or u_t^f. The broken vertical lines are 95 percent posterior intervals. The lower right panel shows the coefficient estimates for RNI_t from a quantile regression model using the draws of u_t^b as dependent variable.

where $\boldsymbol{y}_t = (ne_t, y_t)'$ and y_t is the log-level of GDP, $\boldsymbol{x}_t = (1, tr, \boldsymbol{y}'_{t-1}, \dots, \boldsymbol{y}'_{t-p})'$ a vector containing a constant, a quadratic time trend, and $p = 4$ lags of $\boldsymbol{y}$, $\boldsymbol{\Phi}$ is a matrix of coefficients, and $\boldsymbol{\varepsilon}_t$ a vector of reduced form errors. The relationship between the reduced form errors and the structural shocks is given by $\boldsymbol{\varepsilon}_t = \boldsymbol{A}\boldsymbol{u}_t$. Because $(\boldsymbol{A}\boldsymbol{u}_t)(\boldsymbol{A}\boldsymbol{u}_t)' = \boldsymbol{\Sigma}_{\boldsymbol{\varepsilon}}$, identification is achieved by assuming that $E(\boldsymbol{u}_t\boldsymbol{u}_t') = \boldsymbol{I}$ and by restricting the elements of $\boldsymbol{A}$ such that the following relationship holds:

$$\begin{bmatrix} \varepsilon_{1,t} \\ \varepsilon_{2,t} \end{bmatrix} = \begin{bmatrix} + & - \\ + & + \end{bmatrix} \begin{bmatrix} u_t^f \\ u_t^b \end{bmatrix} \tag{9}$$

As in EKM, u_t^f is interpreted as a fundamental shock moving the nowcast error and output in the same direction. In contrast, a positive belief shock u_t^b leads to a negative nowcast error. But, because expectations are positive, output rises although not as much as expected.

For estimation, we consider the sample 1975:Q1 to 2019:Q4 and assume a normal prior distribution for $\boldsymbol{\Phi}$ and an inverse Wishart distribution for $\boldsymbol{\Sigma}_{\boldsymbol{\epsilon}}$. A total of 100 thousand draws are obtained from the conditional posterior distributions via Gibbs sampling, and thousand draws are retained for inference after thinning the chain. The prior means and covariances are set according to the popular Minnesota scheme (Litterman, 1986), and we use the algorithm proposed in Rubio-Ramirez et al. (2010) for sign identification (of $\boldsymbol{A}$).

Fig. 8 reports the results. The upper left quadrant shows the impulse responses from the SVAR analysis. Both in qualitative and quantitative terms these results are very similar to those reported in EKM: Following a positive fundamental shock we observe delayed updating via the nowcast error response and a fairly persistent, but gradually trend-reverting, output response. In contrast, a positive belief shock which drives down the nowcast error leads to a more u-shaped output response. I.e., the initial response is positive but considerably smaller than the peak response occurring after roughly eight quarters. Fig. A.11 (Appendix A) shows that our results mirror those in EKM also in terms of the forecast error variance decomposition: shocks to fundamentals account for the largest part of the fluctuations of the nowcast error. They also dominate the short-run fluctuations of output. Still, on longer response horizons, belief shocks account for close to 40% of output fluctuations.

The results reported in the upper right graph are new to this study, and illustrate the RNI together with the estimated belief shock, i.e., the (smoothed) posterior estimates of $\hat{u}_t^b$. The lower left graph reports a histogram of the correlation between these two series, where every draw of $\hat{u}_t^b$ is considered. As clearly seen in these graphs, the RNI has a significant and negative correlation with a positive belief shock. In contrast, the correlation between all the draws of $\hat{u}_t^f$ and the RNI is much weaker and insignificant. Finally, in the lower right graph we use a simple quantile regression to evaluate the relationship between the belief shock and the RNI, and find that the RNI is in particular negatively correlated with the lower tail of the belief shock.

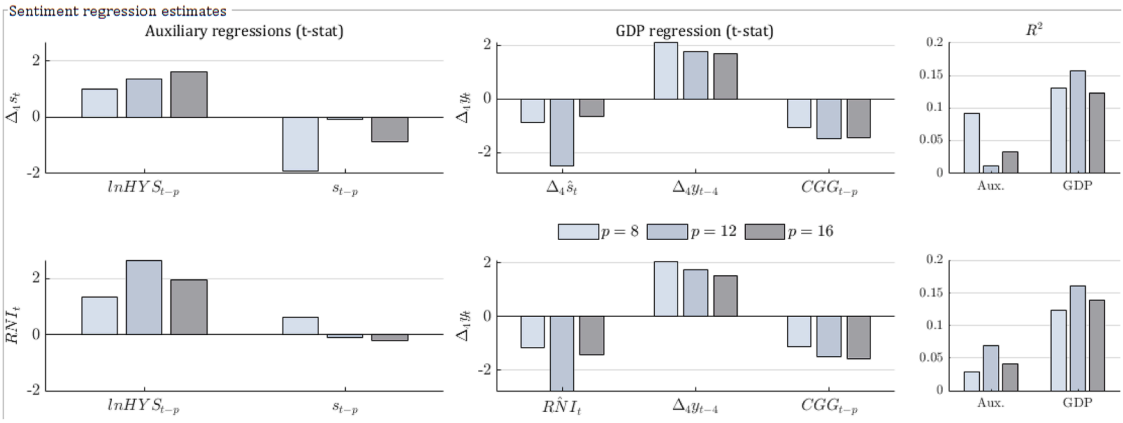


Fig. 9. The panels reports t-statistics and R^2 from estimating (10) using the NLLS estimator. Predictor names are reported on the x-axis and the dependent variable on the y-axis. The different bars for each predictor correspond to a specific lag structure. See the text for details.

In Appendix E we show that all of these findings remain robust when we estimate a reduced-rank SVAR with two common shocks and an expanded set of endogenous variables. Although more complex, this specification allows us to control for fluctuations in forward-looking variables - such as consumer sentiment and asset prices - that help mitigate the non-fundamentalness issues often associated with traditional small-scale VARs (Sims and Zha, 2006; Forni et al., 2017; Beaudry and Portier, 2014). It also delivers a richer characterization of the underlying shocks and their propagation mechanisms, thereby strengthening the credibility of structural interpretation. Notably, in this setting we continue to find a strong negative correlation between the RNI and a positive belief shock, but also document that a positive fundamentals shock lowers the credit-to-GDP gap, whereas a belief shock raises it, which is consistent with a view where beliefs are particularly important for the credit cycle.

4.2. Valuation indicators evidence

LSZ provides an alternative framework for identifying credit-market sentiment more directly from market prices and proxies for expected returns. The econometric framework used is the following two-step regression specification

$$\Delta_4 s_t = \boldsymbol{\theta}' \boldsymbol{z}_{t-p_1} + v_{1,t} \tag{10a}$$

$$\Delta_4 y_t = \beta_1 \Delta_4 \hat{s}_t + \boldsymbol{\gamma}' \boldsymbol{x}_{t-p_2} + v_{2,t}. \tag{10b}$$

In the first-step auxiliary regression (10a) the year-on-year change in Moody's Baa-Treasury credit spread ($\Delta_4 s_t$) is regressed on a vector $\boldsymbol{z}_{t-p_1}$ containing the lagged high-yield bond issuance (HYS_{t-p}), expressed as a share of total bond issuance, and the level of the credit spread (s_{t-p}). As documented in Greenwood and Hanson (2013), these valuation indicators are strongly associated with negative expected returns at horizons ranging from one up to three-years-ahead. In the second step regression (10b) the fitted values from the auxiliary regression ($\Delta_4 \hat{s}_t$) are used to predict business cycle fluctuations conditional on $\boldsymbol{x}_{t-p_2}$. Here, $\Delta_4 y_t$ is computed as year-on-year changes in current quarter (log) GDP and $\boldsymbol{x}_{t-p_2}$ contains lags of $\Delta_4 y_t$ and the credit-to-GDP gap (CGG).[15]

The first row in the left panel in Fig. 9 replicates the analysis in LSZ using our sample and data frequency, and three different lag structures for the valuation indicators and the CGG. As in LSZ, lagged growth is kept fixed at $t-4$ in the second stage regression. Although somewhat imprecisely estimated, the qualitative conclusions from their study hold: Following a period of aggressive credit market pricing, i.e., an elevated level of the high-yield share and narrow (below average) spreads, the component of credit-spread changes that is driven by a reversal of prior positive sentiment ($\Delta_4 \hat{s}_t$) has a negative impact on economic activity.

To analyze to what extent the RNI also contains this sentiment component, we follow the intuition in LSZ and "clean" it for news about fundamentals by re-estimating the model in (10) using the RNI as dependent variable in the first-step auxiliary regression. The second row in the left panel of Fig. 9 reports the results. Consistent with the sentiment-based credit market view, the RNI contains a predictable component positively related to lagged values of high-yield bond issuance and negatively related to lagged credit spreads (at least for lag structures larger than two years), and this predicable component has a negative impact on the business cycle. In terms of significance, we actually find that the HYS is a more accurate predictor of the RNI than of changes in the credit spread while lagged levels of the credit spread tend to matter more for $\Delta_4 s_t$.

To analyze the incremental signal strengths of credit market sentiment, the credit-to-GDP gap and the RNI at business cycle frequency in a more general manner, and at the same time be relatively agnostic about the exact lag structures, we proceed by

[15] We sample the CGG indicator from the Bank for International Settlements (BIS). LSZ consider yearly data for the period 1926 to 2015. Our sample starts in 1985, for which working with annual data yields less than 40 observations. We therefore replicate their main analysis using quarterly observations. We use year-on-year changes to capture business cycle fluctuations and to resemble the annual changes analyzed in LSZ. Similarly, (10) is estimated using nonlinear least squares (NLLS) to take into account the generated-regressor nature of expected returns.

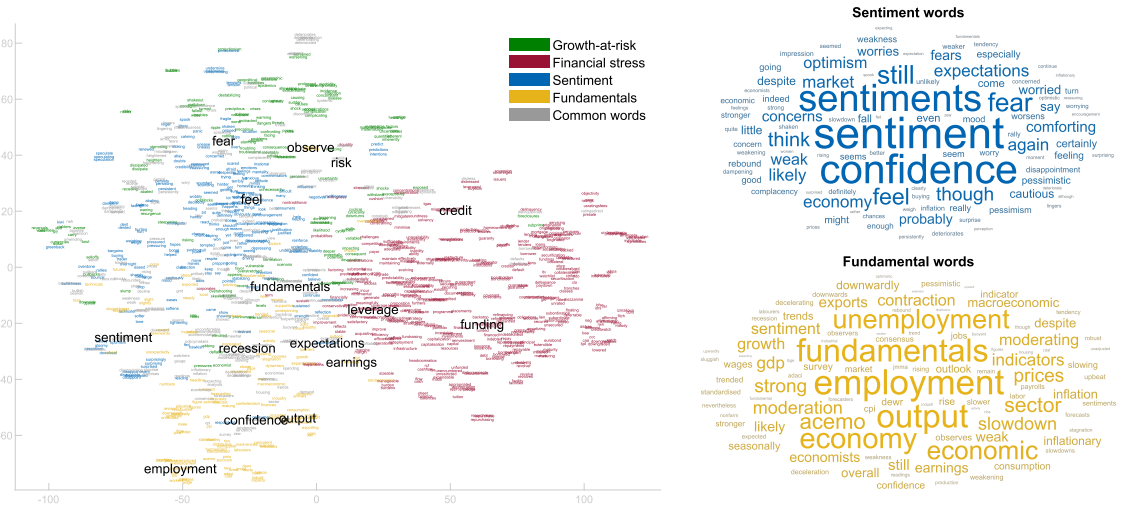


Fig. 10. The left graph reports a scatter plot of the words related to the four concepts listed in the legend. See the text and Appendix B.3 for a description of how the graph is constructed. The graphs to the right are traditional word cloud plots where word size is determined by the cosine similarity scores.

Table 1
The table reports results from regressing similarity measures between $\mathbf{fc}_t$ and sentiment ($\mathbf{S}_t$) and fundamentals ($\mathbf{F}_t$) on the RNI. Newey and West (1987) standard errors in parenthesis. Full sample 1985–2022. Recession dates defined by NBER, but starting one year earlier to prolong the sample.

	Full sample			Recession sample		
	(I)	(II)	(III)	(IV)	(V)	(VI)
S_t	0.482		0.474	0.568		0.759
	(0.132)		(0.133)	(0.234)		(0.250)
F_t		0.175	0.038		0.042	−0.416
		(0.128)	(0.092)		(0.200)	(0.272)
R^2	0.26	0.03	0.26	0.25	0.00	0.32
N	141	141	141	33	33	33

running simple LASSO regressions including as predictors the three different $\Delta_4 \hat{s}_{t,p=.}$ approximations and 8 to 12 quarterly lags of the RNI_t, CGG_t, and $\Delta_4 y_t$. The results from the regularized regression experiments are reported in Appendix D. In short, we document that lags of the RNI are always selected as significant predictors whereas the different $\Delta_4 \hat{s}_{t,p=.}$ approximations and lags of CGG_t are not. Moreover, we show that our earlier finding about the non-Gaussian properties of the conditional distribution of GDP growth holds when using standard quantile regressions and $\Delta_4 y_t$ as the outcome variable.

4.3. Sentiment versus fundamentals in embedding space

As a final experiment, we return to the intrinsic evaluation exercises discussed in Section 2.3. Specifically, we construct two aggregate embeddings intended to capture *sentiment* and *fundamentals*, respectively. The sentiment embedding is defined as the sum of the vectors for *sentiment*, *feel*, *confidence*, *fear*, and *expectations*. The fundamentals embedding is constructed analogously from the vectors *fundamentals*, *observe*, *output*, *employment*, and *earnings*. Using these two composite representations, we pursue two complementary analyses. First, we illustrate their position in vector space using a word scatter plot based on the t-SNE algorithm of Van der Maaten and Hinton (2008). Second, we examine their explanatory power for fluctuations in the RNI using simple regression analysis.

The results of the visualization exercise are shown in Fig. 10, which displays a word scatter plot for the year 2007. The words most similar to the sentiment and fundamentals aggregates are shown in the word clouds on the right-hand side of the figure. Although the two concepts are not entirely orthogonal - some overlap in associated words is expected - the overall composition aligns well with economic priors and reveals a clear semantic distinction. The t-SNE plot reinforces this pattern: sentiment-related terms cluster in a well-defined region of the space, distinct from the cluster formed by fundamental terms. Furthermore, the sentiment cluster lies considerably closer to the growth-at-risk terms than the fundamentals cluster, suggesting a tighter semantic link between sentiment-laden language and the underlying GaR embedding.

Table 1 reports the results from the second-stage regression analysis. Using the two aggregate embeddings defined above, we compute the time-varying cosine similarity between each aggregate - denoted $\mathbf{S}_t$ for sentiment and $\mathbf{F}_t$ for fundamentals - and the monthly $\mathbf{fc}_t$ (financial conditions) embedding. These similarity measures are then regressed on the RNI. For both the full sample, and a sample focusing only on recession episodes, the $\mathbf{S}_t$ is positive and highly significant, indicating that periods in which the financial conditions embedding lies closer to sentiment-laden language coincide with higher values of the RNI and elevated recession risk. In contrast, we find little evidence that the fundamentals similarity $\mathbf{F}_t$ exhibits a comparable relationship with the RNI. When both measures are included jointly, the sentiment measure remains strongly significant, while the fundamentals measure continues to show limited explanatory power.

5. Conclusion

A large literature emphasizes the importance of financial conditions for understanding adverse growth outcomes and growth-at-risk. However, the potentially nonlinear nexus between financial condition indicators and the conditional distribution of GDP growth has recently been challenged, and it is still also debated to what extent fluctuations in financial conditions can best be understood as shocks to fundamentals or more related to behavioral mechanisms and sentiment.

In this article we show how one can use textual economic news combined with a shallow Neural Network to construct an alternative financial conditions indicator based on word embeddings. By design the index, which we label the Risky News Index (RNI), associates growth-at-risk to news about credit, leverage and funding. We then test the index's predictive performance for growth-at-risk and dissects it informational content.

Using quarterly U.S. GDP growth and a parametric skewed Student-t model, a standard modeling approach in this literature, we document that the RNI significantly affects the evolution of the growth distribution's shape. In an extensive out-of-sample forecast analysis the RNI outperforms commonly used financial condition indicators and factors summarizing the developments in real indicators. The performance of the RNI is particularly strong at the one-year-ahead forecasting horizon and during the Great Financial Crisis. For this horizon and episode, growth-at-risk predictions using conventional predictors are relatively uninformative whereas predictions from the RNI suggest a severe recession.

To dissect the informational content of the derived index we link it to structural fundamentals and beliefs innovations, as well as credit market valuation indicators, and document that it is significantly correlated with shocks to beliefs and expected returns. Consistent with this, intrinsic evaluation experiments on the underlying word embeddings suggests that fluctuations in financial conditions and the RNI are more associated with sentiment-laden language than fundamentals. Furthermore, the predictive power of the RNI surpasses that of sentiment derived from credit market valuation indicators, suggesting an avenue for future research where media coverage on financial conditions and growth-at-risk potentially have independent and adverse effects on expectations about the current state of the economy.

Our main results are robust along a number of dimensions, and the proposed methodology outperforms simpler NLP methods linking growth-at-risk to financial conditions, such as count- and Boolean-search-based methods commonly used in economics. This suggest that the word embedding approach might also be useful in other settings where learning word associations from written texts is relevant.

Declaration of competing interest

The authors declare the following financial interests/personal relationships which may be considered as potential competing interests: Paul Labonne reports financial support was provided by Research Council of Norway (NRC 315008). Leif Anders Thorsrud reports financial support was provided by Research Council of Norway (NRC 315008). If there are other authors, they declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Supplementary material

Supplementary material associated with this article can be found, in the online version, at 10.1016/j.jedc.2026.105331

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